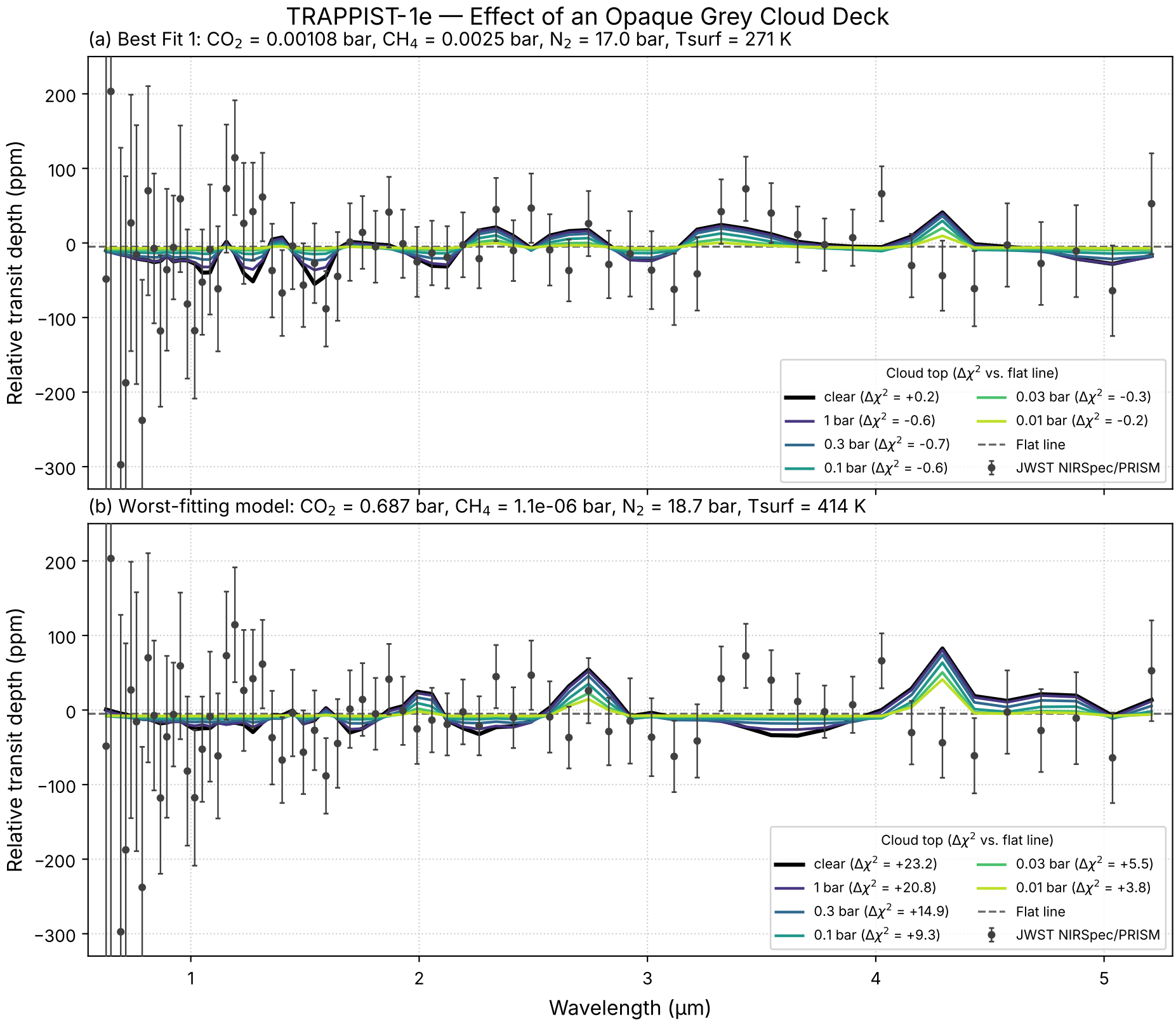
TRAPPIST-1e — Effect of an Opaque Grey Cloud Deck
(a) Best Fit 1: CO2 = 0.00108 bar, CH4 = 0.0025 bar, N2 = 17.0 bar, Tsurf = 271 K
(b) Worst-fitting model: CO2 = 0.687 bar, CH4 = 1.1e-06 bar, N2 = 18.7 bar, Tsurf = 414 K
Relative transit depth (ppm)
Wavelength (μm)
Cloud top (Δχ² vs. flat line)
clear (Δχ² = +0.2)
1 bar (Δχ² = -0.6)
0.3 bar (Δχ² = -0.7)
0.1 bar (Δχ² = -0.6)
0.03 bar (Δχ² = -0.3)
0.01 bar (Δχ² = -0.2)
Flat line
JWST NIRSpec/PRISM
clear (Δχ² = +23.2)
1 bar (Δχ² = +20.8)
0.3 bar (Δχ² = +14.9)
0.1 bar (Δχ² = +9.3)
0.03 bar (Δχ² = +5.5)
0.01 bar (Δχ² = +3.8)
Flat line
JWST NIRSpec/PRISM